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Principal Component Analysis

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PLAN

- Introduction
- **A Step-by-Step Explanation of Principal Component Analysis (PCA)**
- Results interpretation
- Conclusion

1. Introduction

Principal component analysis, or PCA, is a dimensionality-reduction method that is often used to reduce the dimensionality of large data sets, by transforming a large set of variables into a smaller one that still contains most of the information in the large set.

the idea of PCA is simple **— reduce the number of variables of a data set, while preserving as much information as possible.**

A Step-by-Step Explanation of Principal Component Analysis (PCA)

STEP 1: STANDARDIZATION

The aim of this step is to standardize the range of the continuous initial variables so that each one of them contributes equally to the analysis.

Mathematically, this can be done by subtracting the mean and dividing by the standard deviation for each value of each variable.

$$z = \frac{value - mean}{standard\ deviation}$$

Once the standardization is done, all the variables will be transformed to the same scale.

STEP 2: COVARIANCE MATRIX COMPUTATION

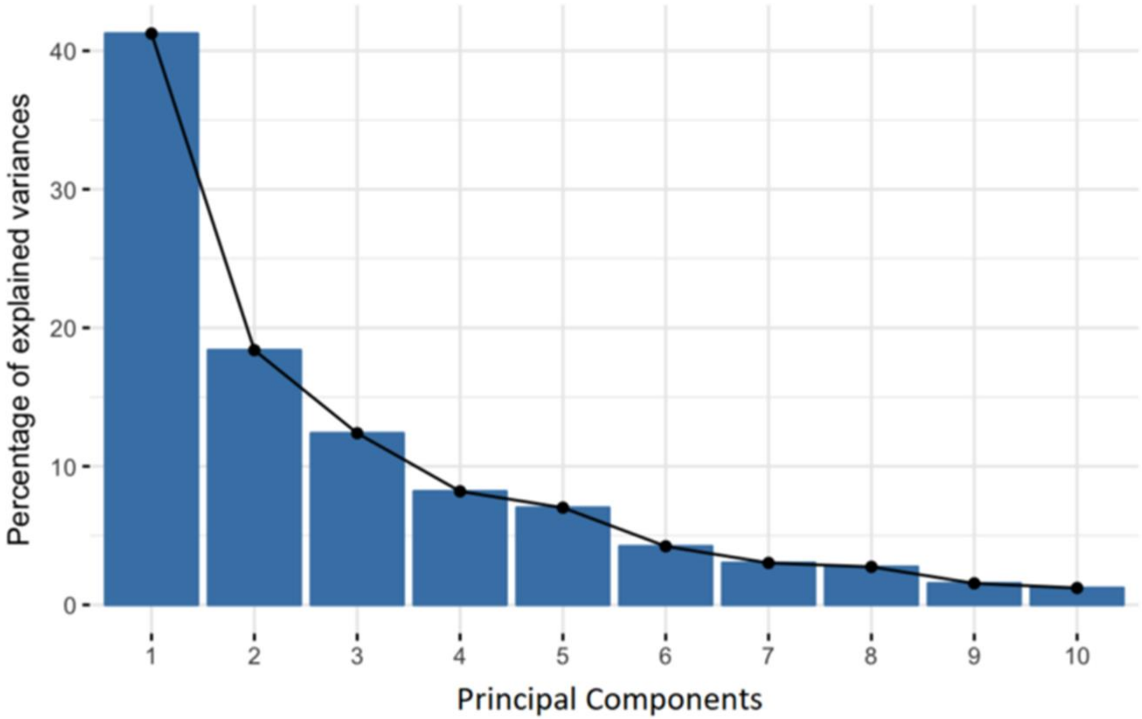
The aim of this step is to understand how the variables of the input data set are varying from the mean, or in other words, to see if there is any relationship between them.

The covariance matrix is a $p \times p$ symmetric matrix (where p is the number of dimensions) that has as entries the covariances associated with all possible pairs of the initial variables. For example, for a 3-dimensional data set with 3 variables x, y, and z, the covariance matrix is a 3×3 matrix of this from:

$$\begin{bmatrix} Cov(x,x) & Cov(x,y) & Cov(x,z) \\ Cov(y,x) & Cov(y,y) & Cov(y,z) \\ Cov(z,x) & Cov(z,y) & Cov(z,z) \end{bmatrix}$$

STEP 3: COMPUTE THE EIGENVECTORS AND EIGENVALUES OF THE COVARIANCE MATRIX TO IDENTIFY THE PRINCIPAL COMPONENTS

Principal components are new variables that are constructed as linear combinations or mixtures of the initial variables



By ranking your eigenvectors in order of their eigenvalues, highest to lowest, you get the principal components in order of significance.

After having the principal components, to compute the percentage of variance (information) accounted for by each component, we divide the eigenvalue of each component by the sum of eigenvalues.

STEP: RECAST THE DATA ALONG THE PRINCIPAL COMPONENTS AXES

This can be done by multiplying the transpose of the original data set by the transpose of the feature vector.

$$FinalDataSet = FeatureVector^T * StandardizedOriginalDataSet^T$$

Remarque

⇒ For Standard ACP

The **variance-covariance** matrix becomes the correlation matrix

The points are projected onto the plane formed by the first two factorial axes.

4. Interpretation of projections

To simplify the interpretation:
we call - The column vectors the variables.
- The Line vectors the observations

Principal plane

This is the plane closest to the statistical units. Each point is placed on the axes using its coordinates.

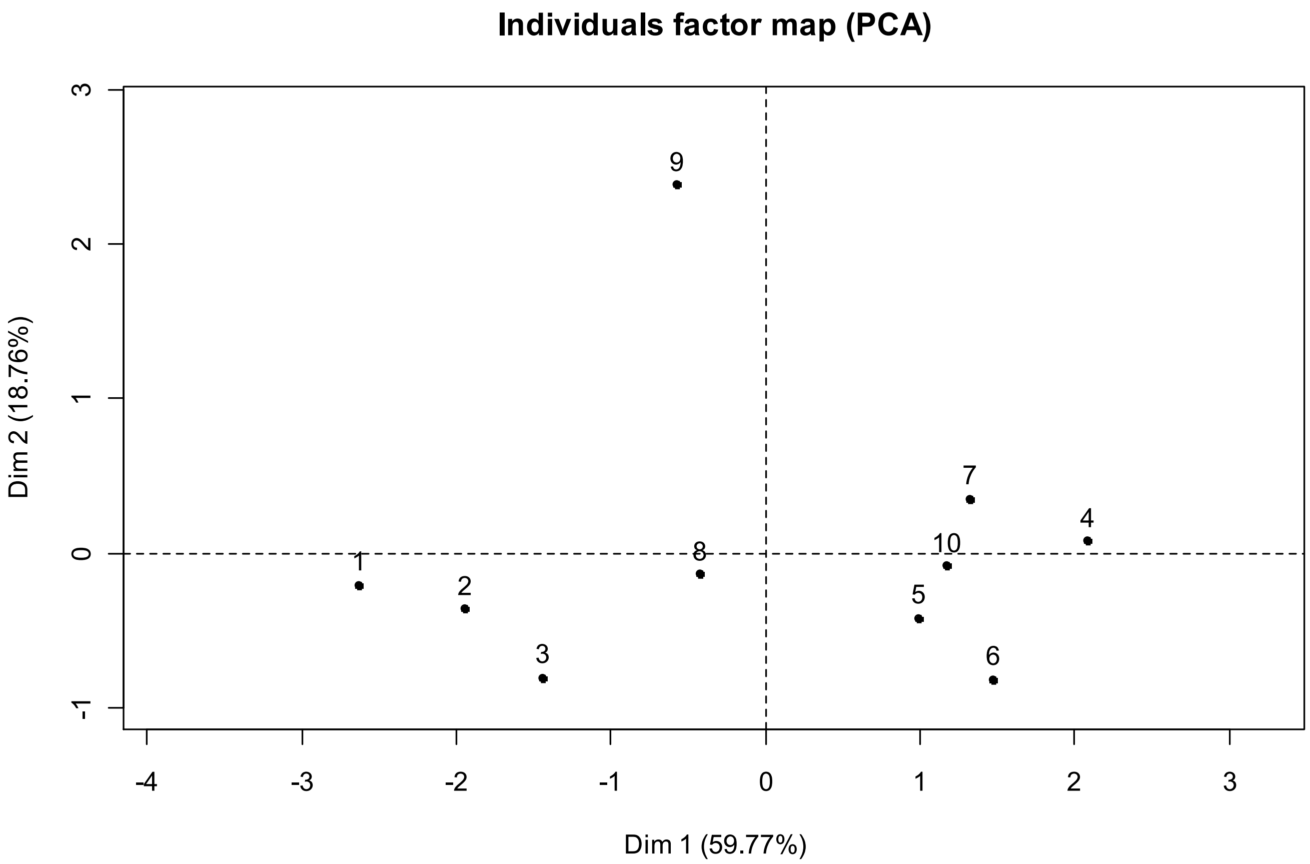
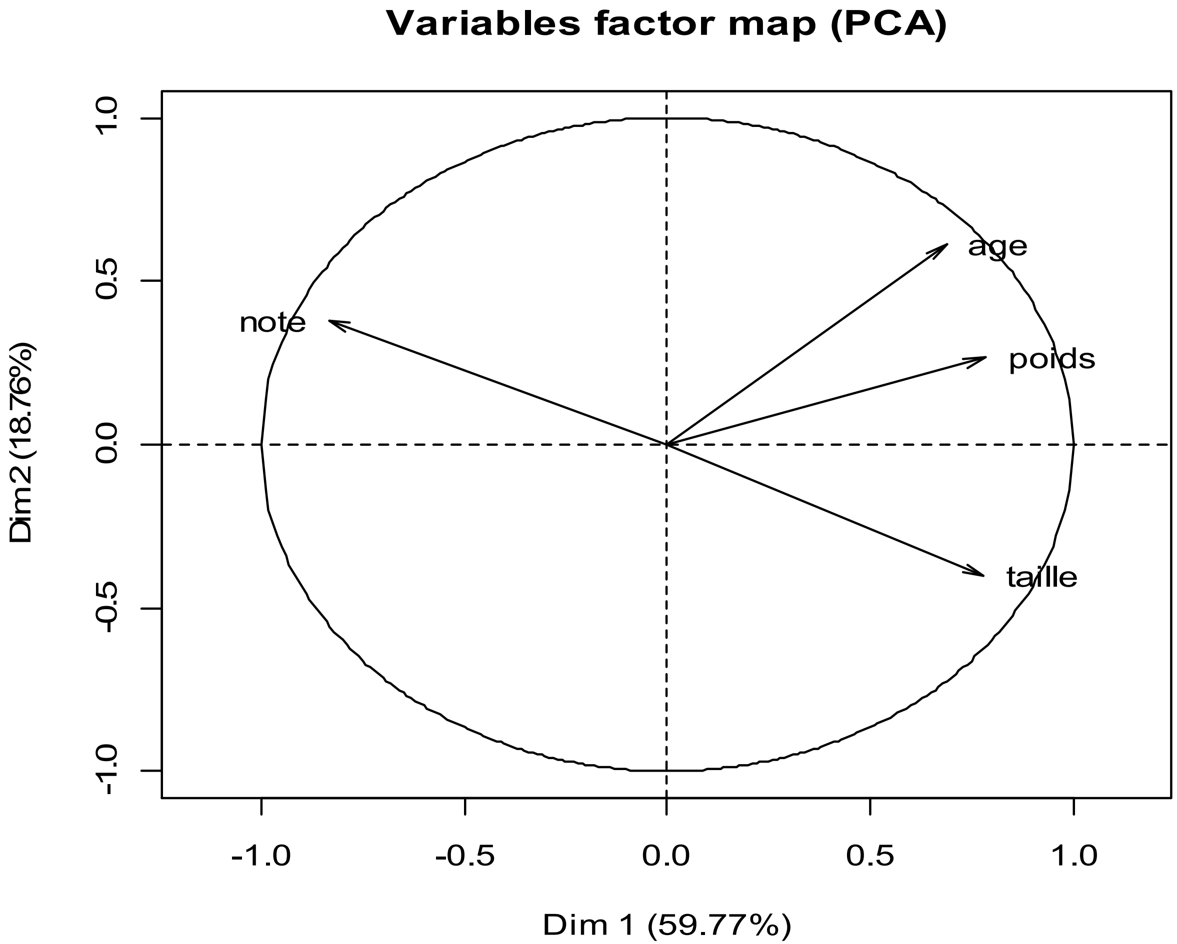


Figure 2.2 : ***plan principal 1 X 2***

A proximity between the projections of two observation points

is **interpreted** as a **similar behaviour** for the q variables

Correlation circle

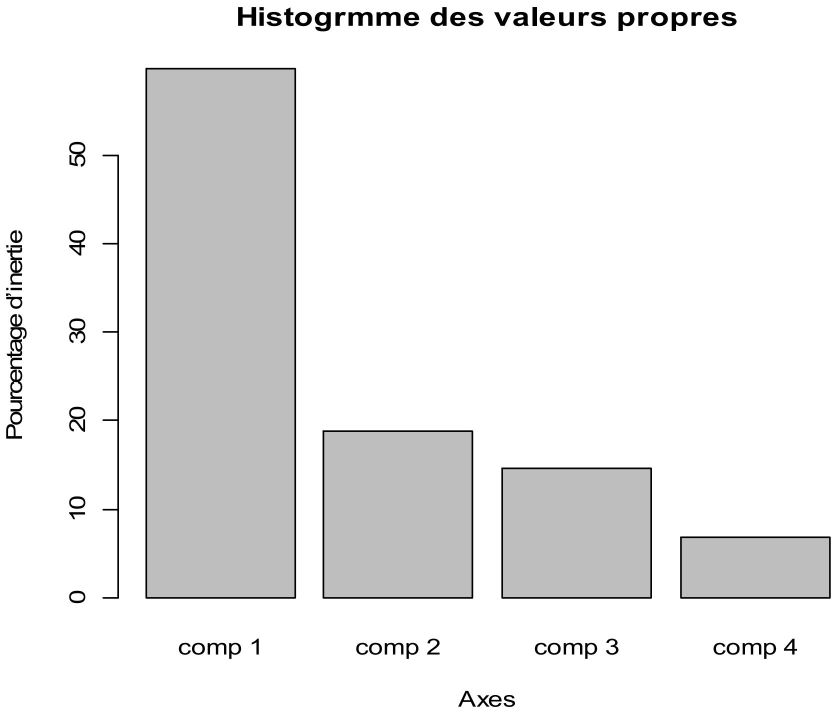


A proximity between two variable points means that the two variables **are correlated.**

The correlation is even more significant when the points are further away from the origin.

Remarque

	eigenvalue	percentage of variance	cumulative percentage of variance
comp 1	2.3908988	59.772471	59.77247
comp 2	0.7503114	18.757786	78.53026
comp 3	0.5844061	14.610152	93.14041
comp 4	0.2743837	6.859591	100.00000



There is no general rule to limit the number of axes considered.
In this example, retaining these three principal components keeps about 93 %of the squares of the distance between the statistical units

The quality of representation

	Dim.1	Dim.2	Dim.3	Dim.4
1	-2.6383478	-0.20304035	-0.10408766	-1.04443266
2	-1.9434521	-0.35783346	0.31559230	0.34952125
3	-1.4422179	-0.80252677	0.59077968	0.48620265
4	2.0830449	0.07837172	1.20080516	-0.19195868
5	0.9867493	-0.42008337	0.29589940	0.05285961
6	1.4737035	-0.81638306	0.06130029	-0.55456712
7	1.3169524	0.35329157	-1.45426142	-0.40902793
8	-0.4313990	-0.13555130	-1.24948767	0.67416087
9	-0.5711642	2.38554456	0.41285443	0.07121803
10	1.1661308	-0.08178953	-0.06939452	0.56602398

	Dim.1	Dim.2	Dim.3	Dim.4
1	0.85896712	0.005087181	0.001336937	0.134608759
2	0.91523525	0.031027512	0.024134475	0.029602762
3	0.62849993	0.194608831	0.105461656	0.071429586
4	**0.74503375**	**0.001054623**	**0.247584676**	**0.006326947**
5	**0.78490790**	**0.142257783**	**0.070581879**	**0.002252439**
6	**0.68955164**	**0.211609196**	**0.001193086**	**0.097646079**
7	0.41879097	0.030138644	0.510672088	0.040398300
8	0.08382389	0.008275933	0.703191313	0.204708859
9	0.05268061	0.918975625	0.027524715	0.000819047
10	0.80381949	0.003954207	0.002846520	0.189379787

unit sta : $\cos^2\theta_1 + \cos^2\theta_2 = \cos^2\theta$

4 : $0.745 + 0.001 = 0.746$

5 : $0.785 + 0.142 = 0.927$

A statistical unit is well represented if the sum of the square cosines is very close to 1

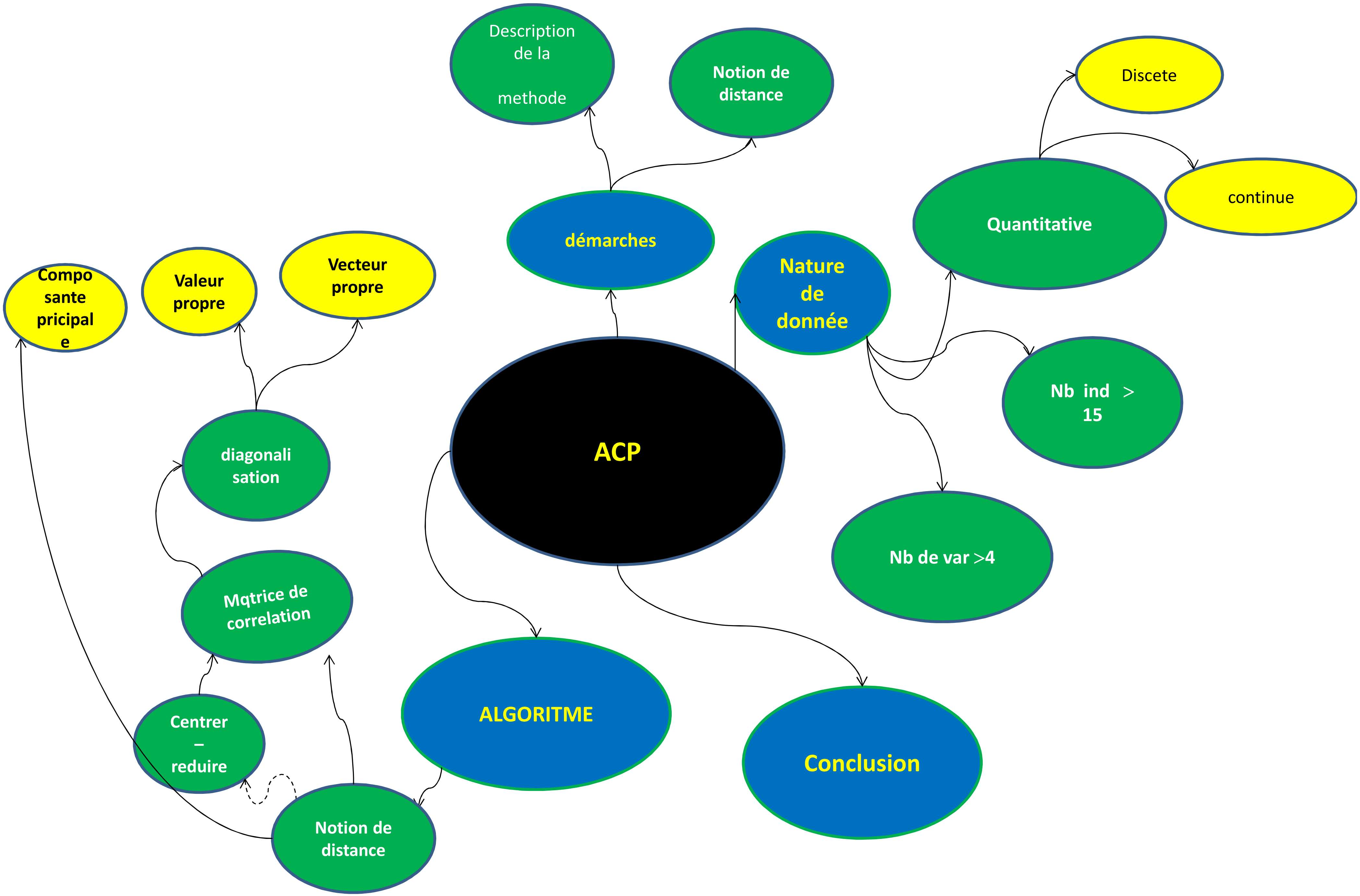
4. Conclusion

✓The principal components keep the same properties of the initial data.

✓ Principal component analysis **on reduced centered** data is used in most cases

✓ this method is generally used for the description of data.

✓collaboration with a specialist is an essential condition for making a **correct interpretation** and reaching **relevant conclusions.**



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